Project Sentinel · Official Submission Manifest

Cryptographic Package Verification & Component Functional Index

Kempower Challenge · LahtiES Energy Tech Hackathon 2026 (LUT University Lahti / CrazyTown)

- **Package Archive**: `Team3_Project_Sentinel_Submission.zip` (also accessible as `submission.zip`)
- **Total Deliverable Files**: 180 (including this manifest & its companion PDF)
- **Total Uncompressed Volume**: 30.16 MB (30,156,963 bytes)
- **Zero-CJK Rule Compliance**: 100% Pure Professional English Verified (`[\u4e00-\u9fff]` match count = 0)
- **Deterministic Verification**: `python 2_The_Demonstration_Part2/verify_all.py` (29 checks passing in < 0.5s)
- **Revision note**: `One_Page_SLA_Specification` restated as a true single A4 page (v3); its full derivations moved to `SLA_Specification_Annex_Full` (rows 177-178)
- **Automated Test Coverage**: 462 Pytest + 106 E2E = 568 total automated tests (100% pass rate)

Deliverable Summary by Pillar

- `root/` (Top-level documentation & manifest): 2 files (607.67 KB)
- `1_pitch_and_story_2029/` (Saturday Deliverable 1: 10-Minute Pitch & 2029 Customer Story): 11 files (17.79 MB)
- `2_demonstration_part2/` (Saturday Deliverable 2: The Demonstration (Scenario Day 13.11.2026 Engine)): 107 files (1.92 MB)
- `3_commercial_and_one_page_sla/` (Saturday Deliverable 3: Commercial Model & One-Page Service Agreement): 8 files (982.60 KB)
- `4_energy_and_control_core/` (Challenge Part 1: Energy & Control Core Architecture & Audits): 48 files (7.35 MB)

Complete Cryptographic Inventory

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
1	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.md	6.13 KB	707c66a792dd8a72ac607c496f7ba289ac568127c4e08a25da58de1f540a1c30	The Midnight Blizzard of Lahti: November 13, 2029 customer story narrative
2	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.pdf	113.56 KB	37a34637b367da08667371d01db8506af4b02ff687006df4ea7c59ad09b74453	Publication-quality PDF of the 2029 customer story narrative
3	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.md	44.95 KB	204e2a9c5640f609e12c9908903296354b73091010728986aa57d005de614917	Detailed pitch deck structure and jury defense talking points
4	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.pdf	667.68 KB	34cd27b347fa9a87734a29e831051cf72d6291c832a973f545bb0b46371ca179	Publication-quality PDF of the pitch deck outline and jury talking points
5	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.md	33.13 KB	a6ce68c132c0a78c6f99a236924df6ec03fc1c0b81ff3750eca3d1556de2bbda	Structured PPT speech plan, slide timings, speaker notes, and stage SOP
6	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.pdf	461.51 KB	eebaf2f2515164ce81e2657ed04d785c43112a7866f03fca44fe185ee9565a36	Publication-quality PDF of the structured PPT speech plan and timings
7	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_12pages_v3.pptx	16.06 MB	9582147dc0ea0de3d668521334c68c49b763877ca6c0c3fc0c232973d852b2ec	Official 12-page executive pitch deck (authentic unmodified original release with real GitHub QR code)
8	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.md	14.79 KB	7a684095e1e6e66e6576a33090932b26743cd887ab94013553f7ae91146292e3	Verbatim 10-minute speech script mapped slide-by-slide to the 12-page pitch deck
9	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.pdf	169.58 KB	cddb4826855c89e1b23886d7995e4c298c3beceac0b9d9a8182951844835a55a	Publication-quality PDF of the 10-minute slide-by-slide speech script
10	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.md	24.88 KB	164dacc7c1fa60cb006b04a51b1c1d6b2f4cf719fb0be90ad24d14a05a83d932	Official 10-minute keynote speech script with blizzard opening and jury defense
11	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.pdf	232.94 KB	e95db4592e14b5317e921c0b016308ede899303804ca0168562e55f0d29219a7	Publication-quality PDF of the official 10-minute keynote speech script

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
12	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.md	6.61 KB	45e53483250d8a88c5b8489313fc29391b09cf6fa0270a9ad1f2cd49841a8b44	Master demonstration guide and scenario day (Friday 13.11.2026) autonomous run proof
13	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.pdf	152.15 KB	979884f995d461cde86dfdeb1f7b817644d7f1fc938ef07b13c90d14b31f21b1	Publication-quality PDF of the master demonstration guide and scenario day proof
14	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa895e7d1f7c528387695c6d1068bbd8b46a8	Deep dive script for thermal sub-zero trip demonstration
15	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	a2d06a42b1cac3973c4c0fda1d41689e544e59511c7bd9b3b898b6745403a00d	Publication-quality PDF of the thermal sub-zero trip demonstration script
16	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0003eac9c1d887c07c5ddc7adc7776055fc	Verbal demonstration speech script for the 6 failure scenarios
17	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.pdf	144.99 KB	d842df14bc3db5cf706d298ab9e8b1fd94f0e7cfef251b0698ca2efda80e3dc5	Publication-quality PDF of the 6-scenario demonstration speech script
18	2_demonstration_part2	2_The_Demonstration_Part2/app.py	32.36 KB	912a1c2e7c073e003ba49a57c8cf9171aa0bf5c7e409837f64380adb3e240735	Interactive 4-View Streamlit Command Center and Executive Cockpit
19	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/.streamlit/config.toml	235 B	34824b6fab69a5c93cd790e1c9180c615cc05f27601f2694801a80479da57bae	Scenario demonstration and interactive replay component
20	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.md	2.50 KB	a8adfad81f84c4e02bd6a7611861892b0d6cce666611a35be83c1cbd3f8c454a0	Scenario demonstration and interactive replay component
21	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.pdf	96.94 KB	6d414d6725b4b713b31479886f2846723000bce562a3373cad47986cf72fba2c	Publication-quality PDF companion document: README.pdf
22	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa895e7d1f7c528387695c6d1068bbd8b46a8	Scenario demonstration and interactive replay component
23	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	f7a5329220a02c862c003c2438cbb3e3f8066c819efe3676cd78b010bad8db71	Publication-quality PDF companion document: SPEECH_OVERHEAT_SCENE.pdf
24	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0003eac9c1d887c07c5ddc7adc7776055fc	Scenario demonstration and interactive replay component
25	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.pdf	144.99 KB	18793eaee8adeefec0aa475598243e5f8e886008e473473cd17bffb6846f7174	Publication-quality PDF companion document: SPEECH_SIX_SCENES.pdf
26	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/app.py	9.78 KB	7d244b2243656319f1f1cccff8b98429d9c40298099f5fca37e38f8d519b545c	Scenario demonstration and interactive replay component
27	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/1_Night_Replay.py	16.84 KB	53a1fa5ef00026ec35a7af4aeebb64decbe17e647fcb4e65738a626522108df4	Scenario demonstration and interactive replay component
28	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/2_Outage_Replay.py	15.19 KB	2af9ddec95691b1e0fbe1eedb1c8c0045232ca0ae16ac2bc4d033194d8e447e0	Scenario demonstration and interactive replay component
29	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/3_Price_Spike_Replay.py	3.16 KB	6ccbed7c0a862a8510c4b1fd8a3311d5b20f51abb2a5fc364ceac66e02331b10	Scenario demonstration and interactive replay component
30	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/4_Curtailment_Replay.py	3.14 KB	bbb06054f14053392b36b7aebccc51b0e360c7d18c9a995f7ea0b0744417e65d	Scenario demonstration and interactive replay component

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
31	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/5_Cable_Fault_Replay.py	3.22 KB	af44344ba7e4cf3c06acba2b351c6c5515f4d9fb5c7697684a75fa6de289400f	Scenario demonstration and interactive replay component
32	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/6_Commissioning_Replay.py	3.07 KB	1c757b29c09d7cb2ae36dbc503862f428b2359a71a83d29dcb2f783087c21be5	Scenario demonstration and interactive replay component
33	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/requirements.txt	491 B	89fcc60e66fbc706deeaaa9777cc639c62c382ced99f1e22fafcefc3690af312	Scenario demonstration and interactive replay component
34	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/__init__.py	90 B	de00142fb1e50bcd4418d8b2c7263ae5e82720dd1ab086172c34814e49aa5979	End-to-End opaque-box testing test suite component
35	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/common.py	4.82 KB	828699c80d91ea344d6ba0ad9b8c53b2a0a18a20d24cac28e8e8e1404dc09730	End-to-End opaque-box testing test suite component
36	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/test_runner.py	5.79 KB	1dcf1cb66c6df0fb0eaa57333b1dae9181175aae632b822893991dc98303edee	End-to-End opaque-box testing test suite component
37	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier1_feature_coverage.py	20.12 KB	8223a3139b255427e58ffd605b7125fcd90c61f7e3b44b8eb693fbc1977a423d	End-to-End opaque-box testing test suite component
38	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier2_boundary_corner.py	13.35 KB	19f07c20f80a3eafeaaabace23545f3bf87cfb070e5cb37780ea0dbe695ae82d	End-to-End opaque-box testing test suite component
39	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier3_cross_feature.py	6.44 KB	4542d7bc49d5805f984319d350943576be5b9324e8400d20fbc7fc8e8088d4d5	End-to-End opaque-box testing test suite component
40	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier4_real_world_scenario.py	8.69 KB	37576eb3980aa71881606eab7b3668d5198a511a068d38f2b2dcb820cc2748b4	End-to-End opaque-box testing test suite component
41	2_demonstration_part2	2_The_Demonstration_Part2/pytest.ini	42 B	b160d82f7811555051f8966d8e02750a985227c1d022208c784e606946f2eada	Pytest configuration specifying test paths and options
42	2_demonstration_part2	2_The_Demonstration_Part2/requirements.txt	488 B	31e6ed5500590b635780438f395a0d0c677a5792be567f90048d2d853273ce9f	Python dependencies and exact package versions
43	2_demonstration_part2	2_The_Demonstration_Part2/run_s1.py	10.13 KB	e0fef5a26ad049e0890b3d6331705eb13aa1a75d55c1a2f85eb50015599fb430	S1 Bus Depot overnight simulation comparing baseline vs zero-human autonomy
44	2_demonstration_part2	2_The_Demonstration_Part2/src/__init__.py	64 B	1abc659bc8548aaa23a37b1b161f89580478d84e044079e58d9b346ae0fc8dac	Project Sentinel deliverable component: **init**.py
45	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/__init__.py	253 B	92bf037063f5bc20ddfdb80cfd5865e762ef11c4638f9bc1e1cb2325dcb4c62c	Project Sentinel deliverable component: **init**.py
46	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/roi_analyzer.py	10.30 KB	a3d8f7a36b1d8de4156579e8b34406bf33d8a838867e68f88496d008ffb55f70	Project Sentinel deliverable component: roi_analyzer.py
47	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sensitivity_analyzer.py	10.88 KB	ed1f8f98fbaffe9d90c5f8d7e24ea04b33ba9689397870e63fed22c44319aae9	Project Sentinel deliverable component: sensitivity_analyzer.py
48	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/settlement_clearing.py	18.15 KB	515f22d2f0ebf93ea465a86190444aa14a0575d6f43e58f45bc8ceb5bfd2fee6	Project Sentinel deliverable component: settlement_clearing.py
49	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sla_calculator.py	4.41 KB	176b755b5e9395ba85653ba7bf7546c483c8c9dc2dfa4cddcd2544ffdbffdda1	Project Sentinel deliverable component: sla_calculator.py
50	2_demonstration_part2	2_The_Demonstration_Part2/src/core/__init__.py	1.99 KB	3ec1b6a99e8e63783cfaae64afcc4ce63bd66900ca	Project Sentinel deliverable component: **init**.py

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
				0f93331869b49a490adcc4	
51	2_demonstration_part2	2_The_Demonstration_Part2/src/core/anti_fraud.py	9.54 KB	17f392630087311f25aae723ab44f45231cbe81d861c91a2f870637ecac78174	Project Sentinel deliverable component: anti_fraud.py
52	2_demonstration_part2	2_The_Demonstration_Part2/src/core/dpr_matrix.py	29.74 KB	139fffd4616829f39f49a4128ead4cb14168b20e9e0cef1359b1dc3899802648	Project Sentinel deliverable component: dpr_matrix.py
53	2_demonstration_part2	2_The_Demonstration_Part2/src/core/mpc_optimizer.py	21.21 KB	b0a4daee6238ab1d0aea01a9abc70070b328ba7f79129c1ec6901536f284c570	Project Sentinel deliverable component: mpc_optimizer.py
54	2_demonstration_part2	2_The_Demonstration_Part2/src/core/protocols.py	40.56 KB	476f7ef80ba7ea7546a50fbe7a481cb06841301b616f1a44e375544c0d5a5a70	Project Sentinel deliverable component: protocols.py
55	2_demonstration_part2	2_The_Demonstration_Part2/src/core/self_healing.py	40.85 KB	1cfd3757821f355631abbd793ff61b1c40a47aa2eae1b37c1cce166e3699d0aa	Project Sentinel deliverable component: self_healing.py
56	2_demonstration_part2	2_The_Demonstration_Part2/src/core/virtual_replug.py	25.39 KB	a7fb6e1f558e6325d89d1b05c71b68f9ada2966941c196015af15f1c3471f25a	Project Sentinel deliverable component: virtual_replug.py
57	2_demonstration_part2	2_The_Demonstration_Part2/src/edf_packer.py	10.87 KB	2c77ed8343639b39f75386842d1191d6972255d7ea24c2b7b008a6abe8af71c2	Project Sentinel deliverable component: edf_packer.py
58	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/__init__.py	265 B	879039c8c27764d2aa6b76ed9c796c608e6d45ab4eadd224b6a2d59349d20e87	Project Sentinel deliverable component: **init**.py
59	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/data_loader.py	4.82 KB	89cacb4f2df1397b1cb9b3088ad55d59c9d314fe5b5912de5a6c10d9052fc0bf	Project Sentinel deliverable component: data_loader.py
60	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/discrete_sim.py	24.13 KB	ead22a7ccb56c684923ebca0b7c69f7da72902136bc22a8c0dba8161831b4388	Project Sentinel deliverable component: discrete_sim.py
61	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/fault_replays.py	24.68 KB	e38becc1669acbbc35f499822b9ef2f2790f92dce1dbd442a94261ff874b8022	Project Sentinel deliverable component: fault_replays.py
62	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/night_replay.py	15.50 KB	473d1b8de78046cec9b1b8911e7284f6f2e068e2e2bc830fe762164e2b912aaa	Project Sentinel deliverable component: night_replay.py
63	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/outage_replay.py	9.24 KB	57e60806400e5e2d1c0033994a29b6e87f8026830db197a840c10d9268cfe6ad	Project Sentinel deliverable component: outage_replay.py
64	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/scenario_engine.py	16.22 KB	1c12c39430a5c7f74444f7425bae10066d34c3efc42cdd50f9cf69ec121632d8	Project Sentinel deliverable component: scenario_engine.py
65	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/__init__.py	84 B	81d9ba7d47beb44af1384215a904f7fdbbcfb151717582a93e637f75241d585b	Project Sentinel deliverable component: **init**.py
66	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/replay_scene.py	22.61 KB	bd6a0fde23441b7f4af13c623e98fcf723d230999833a8a6d11dcfbe8c35c1f8	Project Sentinel deliverable component: replay_scene.py
67	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/__init__.py	211 B	f96583f042dbc2fc7cab4bf9558fb43c4bc10e2208f569547847d79bc385b0fa	Project Sentinel deliverable component: **init**.py
68	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/paths.py	1.57 KB	7d219ac723250015622e02d913447993ba3df3de36763fd6b8ff82463e924abb	Project Sentinel deliverable component: paths.py
69	2_demonstration_part2	2_The_Demonstration_Part2/tests/analyze_scenario_day.py	3.19 KB	9b8d5c9b34b6655cd2adb199ec55b30d57e8ab9af4e36257036968db10641d8e	Pytest automated unit/integration test module

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
70	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_bess.py	554 B	6c6f5f1532f1b68f4f75b03f32ef62638636112ca56f9153d9654ee289acf208	Pytest automated unit/integration test module
71	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_9.py	374 B	415358dd9fef8a0d4e93348461592a2583bbc68d8d7f887a49115dd0f09c412d	Pytest automated unit/integration test module
72	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_later.py	878 B	bda9828a9aee79bb65f99ca0598f971c7fc6ca4e148a79ea897a7f7b4382eb24	Pytest automated unit/integration test module
73	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s1_power.py	704 B	cb16ddba8d133dedf79ae9942d3770f4f0e1b06e1943332ab61b72f35b8615fe	Pytest automated unit/integration test module
74	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s3_outage.py	875 B	91f87cf90aa8b92cced44c86f0f348c90ad324fb2e98a8b14244bde5d2994ecf	Pytest automated unit/integration test module
75	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_t01_t04.py	404 B	e3ed4d5f76d62a9bb5dc632d71d9b6baf2ff3b6f77b814be4a803f9e0f8cc6c4	Pytest automated unit/integration test module
76	2_demonstration_part2	2_The_Demonstration_Part2/tests/conftest.py	157 B	f1efac1390e32ed1e95fc156828ccb5ae89afd49a4e3efa72ae77ad3a74eaa3c	Pytest automated unit/integration test module
77	2_demonstration_part2	2_The_Demonstration_Part2/tests/edf_simulation.py	3.39 KB	6b1952c1a6548386e8446a7bb7e35bf73f94def94605439956486d86e0af697f	Pytest automated unit/integration test module
78	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_cols.py	454 B	8be91976946d4e3a19421fe3f516d220de816d8b8ea2980ab011533a9c3a7d93	Pytest automated unit/integration test module
79	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_dataset.py	955 B	4d3442437c17b4eb003a322ca67f54248de6fd9cbee7722fe70381d76da17f39	Pytest automated unit/integration test module
80	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_s1_telemetry.py	1.10 KB	f06cdaabdd4f8db9305f2b164e69502d3c6681f29ea5e34e83658bb639cf4c36	Pytest automated unit/integration test module
81	2_demonstration_part2	2_The_Demonstration_Part2/tests/query_scenario_data.py	1.53 KB	a5c848437b5e132665655201603f6ef306e6cd9e20b83f56881fa81090eff404	Pytest automated unit/integration test module
82	2_demonstration_part2	2_The_Demonstration_Part2/tests/simulate_s2_curtailment.py	2.75 KB	b88a98fb37592964b92d3802faa6eea4bfa384ee47ae433200bb5eeb6b0db0e7	Pytest automated unit/integration test module
83	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_discrete_sim_thermal.py	24.76 KB	123976b1868da5328a43f6cbfbe09d882ee28916eb93f0d7790726c96abbde80	Pytest automated unit/integration test module
84	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_m1.py	19.24 KB	3d22b03e00023d1c208f3dd876876b84fc6118681d9d5d1d48269a4f9a44417b	Pytest automated unit/integration test module
85	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_anti_fraud_replug.py	5.19 KB	93b485cc6dfc88184f95686be1c720bbc648b906cde86a58327d30b11a9172b8	Pytest automated unit/integration test module
86	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_commercial_settlement.py	15.60 KB	4c73d7bdd9fec16867f1151098a993c9d88f0dae526244f9e5d35854a92702a1	Pytest automated unit/integration test module
87	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dashboard_runtime_stress.py	14.51 KB	ce3f40c9a5c25b6d60c1baff3dd868d248f0af496e4f2df43b2617ba51fab859	Pytest automated unit/integration test module
88	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_demo_day_simulation.py	19.28 KB	bbb4afbd86d92b222b1ef4cd8ed58a2361a728418ccaf6f74254b85e83798827	Pytest automated unit/integration test module

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
89	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_discrete_sim_engine.py	8.00 KB	c6012da71b0f8dc933557e9447236a87c15985978adf3990edd4c117b93c9d0f	Pytest automated unit/integration test module
90	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dpr_stress_adversarial.py	19.89 KB	666effd97188e1f1534e94ff1fd7e1fae43c908e63876cf6be4af6bb3eefc9dc	Pytest automated unit/integration test module
91	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_e2e_suite.py	45.75 KB	10b7ad1b2b5c69f94687a08ac158bf5f05d83a840fca2b9329f78e407057bfe6	Pytest automated unit/integration test module
92	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_gate2_empirical_verification.py	6.97 KB	c4a1e8f11cc9101874febb8a21b534a47426d44f0a1784ca344f7ad67f43f9f0	Pytest automated unit/integration test module
93	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_github_issue2_29items.py	14.33 KB	243ef14cb284c97473a0a12f0ce3a5933ea2a66887205 58f0c5c12ce39662338	Pytest automated unit/integration test module
94	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_adversarial_probe.py	21.96 KB	879b0152b52209f98b372a0ea5860baa4fde47c4081f2d140e85ac502cb7dc7f	Pytest automated unit/integration test module
95	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_remediation.py	13.75 KB	c408e705ba07c7f3f8aa9998e9bb730a03ac7e614b1dda4560047de55c26cef3	Pytest automated unit/integration test module
96	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_challenger_empirical.py	21.39 KB	3daf44cf881c14f7532e2966fdd5e0e1592eaf106bc24925d6ee0317370ba66c	Pytest automated unit/integration test module
97	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_edge_stress.py	17.97 KB	f7118be5fb53d1be6ab324d9b968d7529447de3c3ee1584ab6059add98734be5	Pytest automated unit/integration test module
98	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_self_healing.py	16.38 KB	ab2808b81adaad14d89ae56bd5c73eb20ad6ac86660fc9ad8835fe72a7e7b9c4	Pytest automated unit/integration test module
99	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_protocols_anti_fraud_adversarial.py	19.76 KB	0370eeef177b3a24cd67b85513649d114f1d5efbe655ea979ab1dcd073b94ce1	Pytest automated unit/integration test module
100	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m3_adversarial_stress.py	16.52 KB	218df08d523ca666d4437d6e0351bd58226ca2b803c93be0fa53eaf487ce2637	Pytest automated unit/integration test module
101	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_mpc_math_fixes.py	5.64 KB	a721eaf0bb90da158a2e37d91ae2c3a0fe0cf1ae8f14542ade3749bafd0e948d	Pytest automated unit/integration test module
102	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_protocols_ocpp_iso.py	14.93 KB	dbdddd0672d752b41e1f95372a68280487e17c68cf06662dbc17df20bfa276d2	Pytest automated unit/integration test module
103	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_remediated_sla.py	5.71 KB	b304876d5bd0d4742c288319e519639df3a5b233d8c479d40eea2e854c26ac73	Pytest automated unit/integration test module
104	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_sla_edge_cases.py	3.77 KB	92732951bf0c93e53350d4c8b28e4b16c2b77b5ca3c960e9cc729a5c20047e22	Pytest automated unit/integration test module
105	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_commercial_adversarial.py	42.05 KB	c3f3871bf19d77315e2a5ab9d9dc3393edd488f313f152a457fd84dd2125a237	Pytest automated unit/integration test module
106	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_core_adversarial.py	43.93 KB	a0838458f9ae65f9184428aea32a842b0b649f54bd92a1e47ebcfabe8e050135	Pytest automated unit/integration test module
107	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_csv_inventory.py	2.94 KB	4c97df1cf3dd36ae7cb8aac3c9ea9b3aa70d2df28b612dc0570f471b94b6feb3	Pytest automated unit/integration test module

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
108	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_empirical_claims.py	7.84 KB	c22ca88a0dd01ca42b8601e615e75a3f3e9b838177b31cb7083d7868e3d05c26	Pytest automated unit/integration test module
109	2_demonstration_part2	2_The_Demonstration_Part2/tools/check_demo_artifacts.py	3.12 KB	7f35cdc91661faece79eaf71b2536df05328321d7a9bf1bb9d012d9fbd9b82d5	Engineering verification and artifact analysis utility
110	2_demonstration_part2	2_The_Demonstration_Part2/tools/cjk_audit.py	1.41 KB	3ba5b8cd1979b16a15df80afa705f19e34a5ea8184e380708b3b822b9de0b286	Engineering verification and artifact analysis utility
111	2_demonstration_part2	2_The_Demonstration_Part2/tools/export_deck_ledger.py	39.59 KB	a20260946a57c8857add3b5e3d5b70ef580cc7271223e5b67577c4329cc99378	Engineering verification and artifact analysis utility
112	2_demonstration_part2	2_The_Demonstration_Part2/tools/generate_one_page_docx.py	9.72 KB	7523092e099ad3555eb3385e39071b5d80790c22ff813f8cf7112cb98969688f	Engineering verification and artifact analysis utility
113	2_demonstration_part2	2_The_Demonstration_Part2/tools/md_to_pdf.py	5.74 KB	02a2f804faa6655534301dcf6639ba25ed6635834c85381bc706efcd8c8fdfc5	Engineering verification and artifact analysis utility
114	2_demonstration_part2	2_The_Demonstration_Part2/tools/package_and_manifest.py	16.72 KB	e9378f3f62e06053ea2e5151362afa2083e5b607cce4032574652f695954818a	Engineering verification and artifact analysis utility
115	2_demonstration_part2	2_The_Demonstration_Part2/tools/s1_multicabinet_sim.py	12.12 KB	ea9c9c403c849dd14cecfb3ea9435c29120477dff24cd7ff12c50c9a67584b01	Engineering verification and artifact analysis utility
116	2_demonstration_part2	2_The_Demonstration_Part2/tools/stage_submission.py	8.10 KB	42887b990fb02a8092f88eef9791cd0add5a753144f75d7624803a7a0635e15f	Engineering verification and artifact analysis utility
117	2_demonstration_part2	2_The_Demonstration_Part2/tools/verify_submission_cjk.py	2.33 KB	b142ee1ac6b2c129459ca444d7e79305dbc45a1c2cf4653126f3ee38252840a4	Engineering verification and artifact analysis utility
118	2_demonstration_part2	2_The_Demonstration_Part2/verify_all.py	6.76 KB	f8d05f4dd439d48773ad4aafdd434650a45d4435bf6136fc9f549285a01368c4	Deterministic CLI verifier executing 29 validation checks across 45 CSVs in < 0.5s
119	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.md	4.97 KB	61d6db3b85b684af95aee906ffaa230fd4c428000d823b456e57ec888f5ed645	Comprehensive business model: customers, Certainty-as-a-Service (CaaS), and 20.76 FTE labor waterfall
120	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.pdf	116.12 KB	11310f8f3f93b6e7c1e63aab4ee6b2a37e845e9f7be396fcb52b2c84a29fe210	Publication-quality PDF of the commercial business model and CaaS economics
121	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.md	7.17 KB	a157dede368591551ab21e23c945a6953f6e93809f4e5ae6cae9a3d565ca110c	True one-page CaaS SLA summary: guaranteed SLOs, autonomous authority matrix, penalties and MRDT physics gate (markdown twin of the 1-page PDF)
122	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.pdf	117.00 KB	9740195c69d0c25c95e79495dd29d19d7a522ae8e56a2664eb291a6992966f3b	Publication-quality PDF, exactly 1 A4 page: promised SLOs, what the system decides alone, accountability, caps and execution seal
123	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/Team3-One_Page-Autonomous_Charging_Services_Agreement.docx	38.17 KB	b20b09a2fae74d3ba8799efb1bb2d425124e6d44e51eae69656f63c8b1da62cc	Commercial one-page Autonomous Charging Services Agreement (Microsoft Word format)
124	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.html	11.08 KB	a92cb6b5c7427ea02d216cd5d97ea4cfd157cc43f0c8af9cfd05ada0a999b60d	Rendered one-page SLA agreement strictly formatted to 1 A4 page
125	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.md	6.58 KB	254e3961995d99d9d151b2db2f824a852659df918809bb6ab7506a4a5ee23af7	Markdown source of the one-page SLA agreement
126	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.pdf	148.54 KB	95eed0f121af5cec93d8a79fa224c87b3d96a85e81c47d1666f4490f5e29347e	Exact 1-page A4 publication printout of the Autonomous Charging Services Agreement

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
127	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.md	3.72 KB	be16e96cf571ffabb8f728d489dc26d55ef9e668bc9dca41c8b2b0b971b74fdb	Challenge alignment, competition boundaries, and judging rules reference
128	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.pdf	152.82 KB	7e4a4bffbfc04333cc9ec1804ceca04ef2eb6d900232d997066eac6eb07df413	Publication-quality PDF of the competition boundaries and judging rules
129	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.md	17.15 KB	1804e340c355504f911c24279dc490bceca22cc65617baaee3d39d988c81de14	Single source of engineering truth and system technical blueprint
130	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.pdf	224.51 KB	090beb32e3ac4ef433449771dbbdc87a19d0e41d383984653cb686364c24564f	Publication-quality PDF of the system technical blueprint
131	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.md	7.17 KB	9ee879203249404d8ac5b1c49ea2bee0ad0fb9a2bb428c7c92e4219de829fddf	Forensic audit report verifying all 5 competition evaluation dimensions
132	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.pdf	142.42 KB	c874ee89f4d5fa3fee50d02808552380f3442abcd8ccdfc418def3eae8be77ac	Publication-quality PDF of the forensic audit report
133	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.md	35.02 KB	1847b44b5456b23b5bed36f95b12c5b1b4a6d4f3e89c38d808c79f9b6a09c111	Unified exhaustive technical blueprint, architectural portrait, and file catalog
134	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.pdf	604.49 KB	641617a5779928185dcc7952e6a4cae632351dc13f6a7d3ad2dce340503d015e	Publication-quality PDF of the technical blueprint and file catalog
135	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.md	18.70 KB	4a1df3da519f8b69c763474128e7fccb0bae965ed1d5244bca0efcc9e7ecfc03	Comprehensive zero-human autonomous architecture: L1/L2/L3 intelligence partitioning, offline edge islanding
136	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.pdf	312.53 KB	624c91acfd31cd3a84daaee831dbe9a397da357a45c2d66e823e5144ec5034c1	Publication-quality PDF of the zero-human autonomous architecture design specification
137	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.md	11.67 KB	b51226c0eabfe14f722e204828169e3558e1a40b4ba13d6e06afa4d46c79f2bf	Transparent dictionary of 20 engineering assumptions and introduced artifacts
138	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.pdf	277.50 KB	15011c95aa96a7a08a44a827452e23de560eb7fd87c586f16bb818a0adbdd055	Publication-quality PDF of the 20 engineering assumptions and additions dictionary
139	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.md	13.94 KB	c589f83d68fa7cfcada147a9c1c7e4f8373d318bd45d1b1ac20c926308096e76	16-cell causal attribution matrix for S1, S2, and S5 scenarios
140	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.pdf	466.86 KB	28e2bcf75133c901623a241d2514f81be50c06da504345854131ef244508469d	Publication-quality PDF of the 16-cell causal attribution matrix
141	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.md	8.88 KB	8c1e41f37a71acf3d5591d7c2eaed0d7317803b670bfe04af8069a3ae2fe5842	Demo Day stage defense and jury question bank guide: 00_pocket_cheat_sheet.md
142	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.pdf	157.29 KB	b80f5a55acaebe8c52ca74e83d8ea00abac564d1c3b35ddb715f08b264c9a3df	Publication-quality PDF companion document: 00_pocket_cheat_sheet.pdf
143	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.md	47.69 KB	94bd789edce7e64106fb7bb345c0c8998badab46596fe86a71d41b68fe388921	Demo Day stage defense and jury question bank guide: 01_stage_schedule_and_setup.md
144	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.pdf	433.11 KB	ba9d87b467b95788b5cd11f8d240d47442a5b14f1d92f0f74d3515c834f909a0	Publication-quality PDF companion document: 01_stage_schedule_and_setup.pdf
145	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.md	32.92 KB	35a0133c5902c568cfdc30d30848179cceda5941d2893ddbc35d69408e105a22	Demo Day stage defense and jury question bank guide: 02_team_roles_and_coordination.md

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
146	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.pdf	367.02 KB	76418eee8c8f294af5e8dc184f4072c31f76858e578367d666ad1138999a7f27	Publication-quality PDF companion document: 02_team_roles_and_coordination.pdf
147	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.md	31.10 KB	fbe274474571ad072923fd77e3f4a09f7aaf00346c6cc2676b9ed83988f2747e	Demo Day stage defense and jury question bank guide: 03_judge_personas_and_question_bank.md
148	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.pdf	305.30 KB	d207de84f17dbe8f1d00de5332c97e11d2524d87fded30d18d46c15a1dd83141	Publication-quality PDF companion document: 03_judge_personas_and_question_bank.pdf
149	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.md	48.96 KB	bfde4d27f643c05125d413f31e3906192b155520a32f5ed68c65403856f8f02a	Demo Day stage defense and jury question bank guide: 04_structured_defense_and_crisis_manual.md
150	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.pdf	571.60 KB	0fd7488dc9e11be67b2f9b429bd5fe59cde1b7e169160d59aa2121b8f0280258	Publication-quality PDF companion document: 04_structured_defense_and_crisis_manual.pdf
151	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.md	12.56 KB	d80535b094e780b7ba1b0c5dfe0cf04129f645454910d2c57b8e0079a1712809	Demo Day stage defense and jury question bank guide: 05_live_demo_operational_sop.md
152	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.pdf	288.99 KB	b73b9b7c70cdab5e49a2e63cfceb058b9b6b169cb013d690fe280a0090fedd0f	Publication-quality PDF companion document: 05_live_demo_operational_sop.pdf
153	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.md	12.18 KB	1aafcb5d859dae4de823a21f10d3a0cfdb3a0b0850c8cb3518553b4406169aed	Demo Day stage defense and jury question bank guide: 06_code_proof_and_authenticity_guide.md
154	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.pdf	403.17 KB	f083312c3b9ddcde07fb6bb501de007f1d26309463bb726ac5e1088a890b5cc6	Publication-quality PDF companion document: 06_code_proof_and_authenticity_guide.pdf
155	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.md	35.02 KB	1847b44b5456b23b5bed36f95b12c5b1b4a6d4f3e89c38d808c79f9b6a09c111	Demo Day stage defense and jury question bank guide: PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.md
156	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.pdf	604.49 KB	6079ede685a8fbb7fc6726f3a22838efc265891219e764f019c09dcae4b6fcf2	Publication-quality PDF companion document: PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.pdf
157	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.md	16.53 KB	14e87f49a9ca2ea5871ea0a355b6758117120c670b1fe56b1c8cc3f36b2d26ae	Demo Day stage defense and jury question bank guide: README.md
158	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.pdf	180.53 KB	50c6cb5c462423e5561b04743b08e397981938c69d2a5ddace80608c2c44338d	Publication-quality PDF companion document: README.pdf
159	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.md	20.66 KB	deffb9be6a910a30dc7f277e1bca728dcd253bb1a15aa43f18e2ce45407842c5	Station-by-station energy ledger and 95-site network-wide grid impact proof
160	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.pdf	536.04 KB	7665a53f3e2ae5c06dcb1c4e2733c876deee6c102bf2b6606a3e3e18a39b3c7a	Publication-quality PDF of the station energy ledger and grid impact proof
161	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.md	6.18 KB	571c27eb8738cb0b1a6d1ce3775c6ad1b413acb34473c4e1d7603bfadbf47f27	Official hackathon competition specification: BRIEF.md
162	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.pdf	193.66 KB	f7c60221d65042f16bf900925dc05a81574dd440b4c5e6e88399a21ce66b3a13	Publication-quality PDF companion document: BRIEF.pdf
163	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.md	2.31 KB	11af160f1836588db876ec8d590c705f032476be909dd7dd0afbc4298006ad02	Official hackathon competition specification: BUILD_PLAN.md
164	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.pdf	115.57 KB	1dd1b0aab0daf7fcffc1b38824bbc1e26a4c3b478a81a14062335e597f4c7ac7	Publication-quality PDF companion document: BUILD_PLAN.pdf

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum	Functional Description
165	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.md	3.41 KB	db08a4e7f78a9724bc30a86188fabc432d4bc919ac443858227a198c29d6f967	Official hackathon competition specification: DECISIONS.md
166	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.pdf	156.95 KB	554bbbab1e0c4fae1b9c1e94a5afb9342bc0107baeba5935eec77bf864c50895	Publication-quality PDF companion document: DECISIONS.pdf
167	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.md	5.61 KB	4b3979ec02a20e050d04812cc9ab0f6ece8b85fac0e1348e88ebbfcf12658022	Official hackathon competition specification: DEMO_SCRIPT.md
168	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.pdf	138.88 KB	6f5e5b4a222c981ad735280c7414ab3036ccdb2389c0315eccf40ef28fc4ec72	Publication-quality PDF companion document: DEMO_SCRIPT.pdf
169	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.md	5.51 KB	c6b827e9d0665a249fc563c9e00e0c2a3153374fea6244b8d4d8c2ecc75c2d98	Official hackathon competition specification: JUDGING.md
170	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.pdf	173.19 KB	607dc377f711eace15965db94699bddacfcfcf46b3e141bc0ec68d4d926e8795	Publication-quality PDF companion document: JUDGING.pdf
171	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.md	3.51 KB	76170f3d29259f6c0232486e31fbc944ee97f95a3f556b4e21c4dbe4847ee9d0	Official hackathon competition specification: RULES.md
172	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.pdf	135.55 KB	91cfe99e7d8771911872651b030b2fdc601498e3b95e14cebe35f9f47335c188	Publication-quality PDF companion document: RULES.pdf
173	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.md	5.32 KB	13f6abe5fee18c06ecbc981db851a47a3a766dfc479895ea66e0ef1ec58ad456	Official hackathon competition specification: SUBMISSION.md
174	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.pdf	177.94 KB	8cc46dc94e9ae1ef1a3e003746d70f263d2d39dcbbef168e397de38b2a0c6d57	Publication-quality PDF companion document: SUBMISSION.pdf
175	root	README.md	24.05 KB	d1433f2282812746cfeb884e428e9684c58eb3fa91c1330f4e4020b470794656	Master project documentation, system architecture, 6-scenario benchmark, and quickstart guide
176	root	README.pdf	583.62 KB	59fbbda8adb558f38ad52708fd639a3e65aa96460288e41685819bcebfc00bdf	Publication-quality PDF of master project documentation and quickstart guide
177	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/SLA_Specification_Annex_Full.md	17.42 KB	e49bcc4ce230daf95b57bd46e0b4382f7d48830c6fa1c8fa20acbada6a5f2004	Full-reference annex of the SLA specification: 2029 customer story, tier tables, equations, boundaries (markdown twin)
178	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/SLA_Specification_Annex_Full.pdf	173.60 KB	8f078ee44eb6af61915cd00449662900a5a635a8e31514fe1290a2fa4a2deaa1	Publication-quality 4-page PDF annex carrying the complete derivations behind the one-page SLA specification

Independent Verification Instructions

To independently verify the integrity of all files in this submission, run: ```bash

1. Unzip the official archive

unzip Team3_Project_Sentinel_Submission.zip -d submission_verify/ cd submission_verify/2_The_Demonstration_Part2/

2. Run the deterministic verifier (< 0.5s)

python verify_all.py

3. Run the complete E2E and Pytest test suites

python -m e2e_tests.test_runner pytest tests/ -q ```